

Intensity; in-ten-si-ty; *noun*. 1. Often used ambiguously within resistance training. 2. Is it time to drop the term altogether?

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Received 21 December 2012

Revised 21 December 2013

Accepted 19 January 2013

INTRODUCTION

Recently in *BJSM*, Dr Berkoff¹ highlighted some 'hot topics' in sports and exercise medicine. A variety of topics were covered, all of which were 'hot'. Of particular interest, however, was the fact that Dr Berkoff preceded his article with a *definition* of 'hot'. Within sports and exercise medicine, and indeed in all scientific disciplines, definitions are of great importance. In fact, "The primary advantage of operational definitions lies in the unification of science and the resolution of controversy."² It is the definition and use of a term within a topic that might also be deemed as 'hot' that this editorial attempts to address: *Intensity* in resistance training (RT).

Recent publications regarding RT have attempted to offer clarification on the definition of *intensity*.³⁻⁴ Fisher and Smith³ wrote regarding the use of the term *intensity* within RT suggesting that it is better representative of *effort*, whereas other authors have considered it synonymous with *load*.⁴ Fisher and Smith² are not the first to suggest that the use of *intensity* to refer to *load* in RT is inappropriate. Others have previously attempted to instigate a change in the language used by researchers and practitioners.⁵⁻⁶ Despite previously finding myself supporting this view (that *intensity* is better defined as *effort* and not *load*) regarding the use of the term and having published as such,⁶⁻⁷ further consideration has left me doubting the value of both interpretation of *intensity* as synonymous with *load* or *effort*. Thus, this editorial seeks to ask the readers of *BJSM*, and the wider community involved in RT, to consider the use of the term *intensity* as it stands and whether both sides of the disagreement are defending inappropriate idioms. More specifically, however, it asks, 'Is it time to drop the term altogether?'

IS INTENSITY A LOAD OR AN EFFORT?

As mentioned, in previous publications I have supported the view that *intensity* is representative of *effort* and not of *load*.⁶⁻⁷ The crux of the argument for why *intensity* does not represent *load* in RT, and why *intensity* should instead be considered as a representative of *effort* is as follows.³⁻⁴ The use of *load* implies that two individuals performing the same number of repetitions at the same relative load are working at the same intensity. However, research shows that maximal repetition ranges at relative loads can vary considerably between individuals⁸⁻⁹ and so considering *load* may not offer an appropriate means of understanding the effort put forth during resistance exercise. Based upon these

considerations, only one accurate control of *effort* is possible during specific RT, that of 100%, that is, when the participant reaches maximal effort or momentary muscular failure (MMF).^{5-6, 8-10} For example, assuming two persons' one repetition maximum (1 RM) is 100 kg and *load* was considered as a representative of *intensity*, if one individual performs an exercise with a load of 80 kg (80% 1 RM), and performs one easy repetition with that load, this person is training more *intensely* than another individual who performs a hard set to MMF with 79 kg (79% 1 RM).⁴ Shimano *et al*,⁹ however, showed that rating of perceived exertion (RPE) is similar regardless of load when repetitions are taken to MMF.

Yet, a further part of the argument also made by Fisher and Smith³ against the use of *intensity* to refer to the load used during RT was against the fact that others consider it the common understanding and usage of the term in practice.¹¹⁻¹² It has occurred to me that I and my fellow colleagues have also been following a similar idiom as others¹¹⁻¹² with the use of *intensity* to refer to *effort*. For example, Fisher and Smith³ suggest that "Within the area of cardiovascular or aerobic exercise, *intensity* is *generally considered* to represent the effort required by the body..." the issue here being that, although within that particular modality of exercise, *intensity* is commonly considered synonymous with *effort*, this does not necessarily imply that it is being used correctly in that instance either. For example, Fisher and Smith¹ highlight that it is often represented as a percentage relative to maximum heart rate (HR). However, HR can significantly change in the absence of physical effort from exercise as shown by the significant increases seen in preparing for and engaging in psychologically stressful situations,¹³ suggesting that the two are not synonymous even within cardiovascular exercise.

INTENSITY PROPERLY DEFINED AND USED

The Oxford English Dictionary¹⁴ offers as a definition for the term 'intensity', "The degree or amount of some quality, condition, etc; force, strength, energy; degree of some characteristic quality, as brightness, etc; especially, in physics, as a measurable quantity." A consideration of this definition suggests that I, my fellow colleagues and the rest of the sport and exercise medicine community involved in RT and exercise as a whole are using the term *intensity* incorrectly. More precisely, *intensity* should be used only in one context (and perhaps that should even be avoided altogether for

To cite: Steele J. *Br J Sports Med* Published Online First: [please include Day Month Year] doi:10.1136/bjsports-2012-092127

the sake of clarity). That context is when stating or asking a question regarding the 'intensity OF' something relative to a maximal value. As the dictionary definition suggests, intensity concerns the magnitude or degree of some measureable quantity. Thus, in cardiovascular exercise, it would be more appropriate to speak of the *intensity OF velocity/incline/resistance*, etc during exercise and measure the *intensity OF heart rate/blood lactate/oxygen uptake*, etc. In RT, we could talk of the *intensity OF load* as being the percentage of 1 RM or maximum voluntary contraction that is being used. Or we might talk of the *intensity OF effort* involved during an exercise with the caveat that we can only gain subjective measurement of the sense of effort through RPE, and measurement of motor unit (MU) recruitment in RT provides a physiological variable correlating with effort, with max MU recruitment representing max effort independent of load.^{10 15}

Along this vein of thought, however, we might also consider the differentiation between *sense of effort* and the *physical sensations* associated with exercise, which, considering the above definition of *intensity*, would also be inappropriately labelled as such. Indeed, these phenomena have also been recently called upon to be differentiated appropriately within the articles published in *BJSM*.^{16 17} Smirmaul¹⁶ offers a practical example appropriate to RT in this regard suggesting, "A short maximal voluntary contraction for leg extension, for example, will induce a maximal sense of effort while, initially, other unpleasant sensations will probably be modest. Repeating this maximal contraction several times, however, will increase these unpleasant sensations continuously, whereas sense of effort will always be the same (ie, maximal)." Indeed, I and my colleagues have questioned the use of RPE in RT to determine relative effort.⁶ Shimano *et al*⁹ showed that, when repetitions were continued to MMF, RPE was similar at 60%, 80% and 90% of 1 RM for most exercises; however, RPE was significantly higher during 60% 1 RM squat performed to MMF. *Effort* would have been maximal between conditions considering they were continued to MMF, and thus the different RPE during 60% 1 RM squats may be an artefact of the measure constituting both *sense of effort* and the *physical sensations* associated with the longer duration of RT performed for the legs. Perhaps the use of tools such as those suggested by Swart *et al*¹⁷ to differentiate these two distinct phenomena should also be applied to future RT research. Yet in spite of this, and practically speaking, in order to appropriately control effort in order to examine the impact of other RT variables on these distinct phenomena, only one way of controlling *effort* in research exists and that is having a participant train maximally irrespective of load, that is, to MMF.⁶

CONFUSION IS BORNE FROM AMBIGUITY

As an example of the confusion that an imprecise use of the term intensity can cause while drawing conclusions from research, especially with regard to understanding the impact of *load* or *effort*, I refer to a recent article published in *BJSM*.¹⁸ Kristensen and Franklyn-Miller¹⁸ conducted a systematic review of the use of RT in musculoskeletal rehabilitation in a valuable attempt to provide guidelines as to the optimal manipulation of RT variables for use in populations with musculoskeletal disorders. As such, one of the variables they considered in their interpretation of the studies was *intensity*. One of the musculoskeletal conditions reviewed was chronic low back pain, and two of the studies reviewed claimed to compare what were termed 'high intensity training' (HIT) and 'low intensity training' (LIT) with each other¹⁹ and also with a waiting list control group.²⁰ However, despite *intensity* being the

variable of interest in these studies, no definition of the same was offered, and as such, difficulty is rendered in drawing firm conclusions from the results of both. The manner in which Helmhout *et al*¹⁹ appear to have defined intensity within this study in fact refers to *load*, and as such, the result was that relative *effort* was not controlled. Training for the HIT group in the first study¹⁹ involved 15–20 repetitions during weeks 1 and 2, and 10–15 repetitions between weeks 3 and 12, with a load equal to 35% of the participant's maximal isometric lumbar extension strength. Once the participant could complete more than the maximum number of repetitions, the load was increased by 2.5 kg (no indication of whether the participants were encouraged to continue repetitions past the maximum required when able to, ie, to MMF, was specified). In contrast, the LIT group performed 15 repetitions during the first and second weeks after each test, and 20 repetitions during the 3rd and 4th week after each test using only 20% of their maximal lumbar extension strength as measured in week 1. In the second study, Harts *et al*²⁰ utilised the same programme but increased the load used by the HIT group to 50% of their maximal lumbar extension strength while keeping the LIT group's load the same as previously.

Kristensen and Franklyn-Miller¹⁸ commented that "... in both these above studies, the definition of high-intensity RT was different to the conventional definition of high-intensity RT in a healthy population." Kristensen and Franklyn-Miller appear to be considering the definition of intensity as load as they note that exercise *intensity* of ">70% of 1RM.... can be classified as high-intensity." They also suggest that it is unsurprising that there were no significant adaptations in either of the groups as both were training at a low *intensity* (of load). However, it is unclear whether the low loads used in these studies were indeed responsible for the lack of adaptations and this confusion stems from the use of terminology in the two original studies examined. The *intensity OF load* used by the groups in the studies^{19 20} did indeed differ. But as it is unclear whether either of the groups is trained to MMF, it is likely that *intensity OF effort* also differed and it impossible to know to what degree effort (and thus perhaps also MU recruitment^{10 15}) may have differed between them. Effort will increase with an increase in load assuming all other variables are held constant, yet the loads used in these studies and the degree of difference between groups was rather small (HIT used 35/50% of max strength, LIT used 20% of max strength). In fact, the LIT group may have trained at a relative load similar to the HIT group as the authors noted that even the lowest load that could be used on the lumbar extension machine could not be set low enough to achieve 20% in some participants. In considering the typical repetition ranges that are possible at different loads relative to maximal strength,^{8 9} and the repetition ranges used within this study, it would be more appropriate to conclude that both groups in fact were trained at what could be considered a low *intensity* of both *load* and *effort*. Because the appropriate application of both *load* and *effort* may have importance for producing optimal adaptations, it is perhaps not surprising that the studies found very little differences between both the HIT and LIT groups.

CONCLUSION AND A PLEA TO PEERS

To conclude this editorial, I hope that I have opened up a consideration of the appropriate use of the term *intensity* in the minds of the *BJSM* readers, and I would urge all those involved in RT to consider their current use of the term *intensity*. Intensity refers to the degree or magnitude of a measurable characteristic or variable. If it is not being used in terms of the

exact definition offered by the Oxford English Dictionary, it should be clarified in the instance used, for example, if it is being used as synonymous with *load* or *effort* being that they are the terms most commonly considered. However, despite probably being an unfavourable recommendation for some, I think more appropriate would be the withdrawal of the term altogether. Greater clarity or withdrawal of use of the term would result in far greater lucidity for researchers, practitioners and recreational exercisers alike in reading and interpreting the studies conducted, which often can be a guessing game in this regard as to whether they offer insight as to the optimal manipulation of *load* or *effort*. Appropriate use of this term is something I will personally be striving to achieve in any further publications in order to make interpretation of my work by my colleagues and peers easier, and I offer my apologies for any misunderstanding caused by my previous work in which, although I have clarified intensity in reference to effort, the use of the term was inappropriate nonetheless.^{6 7}

Competing interests None.

Provenance and peer review Not commissioned; internally peer reviewed.

REFERENCES

- Berkoff DJ. What's hot today? Current topics in sports and exercise medicine. *Br J Sports Med* 2013;47:1–2.
- Boring EG. The use of operational definitions in science. *Psychol Rev* 1945;52:243–5.
- Fisher J, Smith D. Attempting to better define 'intensity' for muscular performance: is it all a wasted effort? *Eur J Appl Physiol* 2012;112:4183–5.
- Sakamoto A, Sinclair PJ. Muscle activations under varying lifting speeds and intensities during bench press. *Eur J Appl Physiol* 2012;112:1015–25.
- Carpinelli R, Otto RM, Winett RA. A critical analysis of the ACSM position stand on resistance training: insufficient evidence to support recommended training protocols. *J Exerc Physiol* 2004;7:1–60.
- Fisher J, Steele J, Bruce-Low S, *et al.* Evidence-based resistance training recommendations. *Med Sport* 2011;15:147–62.
- Steele J, Fisher J, McGuff D, *et al.* Resistance training to momentary muscular failure improves cardiovascular fitness in humans: a review of acute physiological response and chronic physiological adaptations. *J Exerc Physiol* 2012;15:53–80.
- Hoeger WWK, Hopkins DR, Barette SL, *et al.* Relationship between repetitions and selected percentages of one repetition maximum: a comparison between untrained and trained males and females. *J Strength Cond Res* 1990;4:46–54.
- Shimano T, Kraemer WJ, Spiering SA, *et al.* Relationship between the number of repetitions and selected percentages of one repetition maximum in free weight exercises in trained and untrained men. *J Strength Cond Res* 2006;20:819–23.
- Carpinelli R. The size principle and a critical analysis of the unsubstantiated heavier-is-better recommendation for resistance training. *J Exerc Sci Fitness* 2008;6:67–86.
- Fry AC. The role of resistance exercise intensity on muscle fibre adaptations. *Sports Med* 2004;34:663–79.
- Willardson JM, Burkett LN. The effect of different rest intervals between sets on volume components and strength gains. *J Strength Cond Res* 2008;22:146–52.
- Kirschbaum C, Pirke KM, Hellhammer D. The 'Trier Social Stress Test'—a tool for investigating psychobiological stress responses in a laboratory setting. *Neuropsychobiology* 1993;28:76–81.
- Oxford English Dictionary 2012. Oxford: Oxford University Press.
- Jungblut S. The correct interpretation of the size principle and its practical application to resistance training. *Med Sport* 2009;13:203–9.
- Smirmaul BDC. Sense of effort and other unpleasant sensations during exercise: clarifying concepts and mechanisms. *Br J Sports Med* 2012;46:308–11.
- Swart J, Lindsay TR, Lambert MI, *et al.* Perceptual cues in the regulation of exercise performance—physical sensations of exercise and awareness of effort interact as separate cues. *Br J Sports Med* 2012;46:42–8.
- Kristensen J, Franklyn-Miller A. Resistance training in musculoskeletal rehabilitation: a systematic review. *Br J Sports Med* 2012;46:719–26.
- Helmhout PH, Harts CC, Staal JB, *et al.* Comparison of a high-intensity and a low-intensity lumbar extensor strengthening training program as minimal intervention treatment in low back pain: a randomized trial. *Eur Spine J* 2004;13:537–47.
- Harts CC, Helmhout PH, de Bie RA, *et al.* A high-intensity lumbar extensor strengthening program is little better than a low-intensity program or a waiting list control group for chronic low back pain: a randomised clinical trial. *Aust J Physiother* 2008;54:23–31.



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Br J Sports Med published online February 12, 2013
doi: 10.1136/bjsports-2012-092127

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